

Therapeutic potential of turmeric in Alzheimer's disease: curcumin or curcuminoids?

Alzheimer's disease (AD) is the most common form of dementia. There is limited choice in modern therapeutics, and drugs available have limited success with multiple side effects in addition to high cost. Hence, newer and alternate treatment options are being explored for effective and safer therapeutic targets to address AD. Turmeric possesses multiple medicinal uses including treatment for AD. Curcuminoids, a mixture of curcumin, demethoxycurcumin, and bisdemethoxycurcumin, are vital constituents of turmeric. It is generally believed that curcumin is the most important constituent of the curcuminoid mixture that contributes to the pharmacological profile of parent curcuminoid mixture or turmeric. A careful literature study reveals that the other two constituents of the curcuminoid mixture also contribute significantly to the effectiveness of curcuminoids in AD. Therefore, it is emphasized in this review that each component of the curcuminoid mixture plays a distinct role in making curcuminoid mixture useful in AD, and hence, the curcuminoid mixture represents turmeric in its medicinal value better than curcumin alone. The progress in understanding the disease etiology demands a multiple-site-targeted therapy, and the curcuminoid mixture of all components, each with different merits, makes this mixture more promising in combating the challenging disease. Copyright © 2013 John Wiley & Sons, Ltd.